

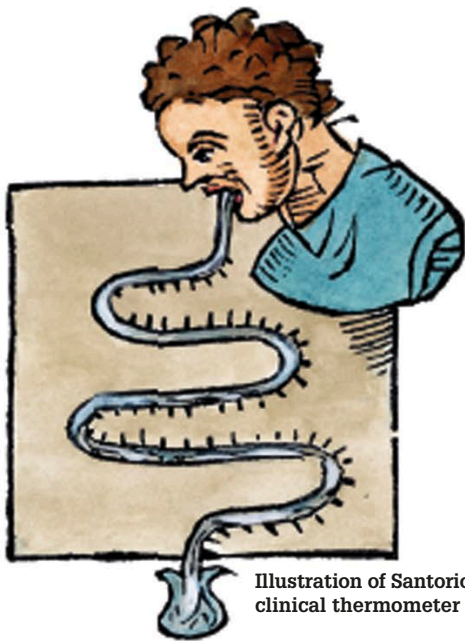


Illustration of Santorio's clinical thermometer

1626

Body temperature

Italian physiologist Santorio was the first person to use a thermometer to measure the temperature of the human body. Santorio was an early pioneer of the study of metabolism—the chemical processes in the body essential for life.

1578–1657

WILLIAM HARVEY

English physician William Harvey proved that the heart pumps blood around the body. He found that the body has a fixed amount of blood that is always circulating. Before this discovery, doctors believed that blood was continuously being made in the liver.

Royal physician

Harvey studied at the universities of Cambridge, England, and Padua, Italy. On returning to England, he became physician to King James I, and tended to victims of the English Civil War (1642–1651).



One-way flow

Arteries carry blood away from the heart and veins return it to the heart. This key discovery of the circulatory system was outlined in Harvey's book *De Motu Cordis* (The Motion of the Heart), where he shows how a one-way valve in a forearm vein prevents the blood from flowing back to the hand.

1625

1630



1629

Fanciful steam engine

Italian inventor Giovanni Branca published a design for a steam engine known as an aeolipile. He suggested it could be used to power pestles and mortars to grind medicinal drugs. It was never built and it is doubtful it would have been of any practical use.

1626

Death by freezing

Sir Francis Bacon, a major English figure in the history of science, is said to have died of pneumonia after trying to see if he could preserve meat by stuffing a chicken carcass with snow. Bacon famously maintained that scientists should prove the truth of their ideas through experimentation.

1. Fire heats water in boiler shaped like a man.

2. Steam is blown through a pipe onto a horizontal turbine.

3. Rotating turbine shifts a series of gears.

4. Weighted arms rise and fall.

Branca's aeolipile, an early type of steam engine

5. Pestle moves up and down.

